SMART INDIA HACKATHON 2026

AI-Powered Criminal Network Analysis System

A Software-Based Solution Proposal for the Ministry of Home Affairs

Problem Statement ID:	SIH26189
Target Organization:	Ministry of Home Affairs (MHA)
Domain:	Blockchain & Cybersecurity / AI & ML
Category:	Software Edition (Purely Software-Based)

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1. Executive Summary & Problem Background

Modern criminal investigations deal with an overwhelming volume of data distributed across unstructured text documents, spreadsheets, and databases. Investigative materials include **First Information Reports (FIRs)**, **Call Detail Records (CDRs)**, **Financial Transaction Logs**, surveillance outputs, and social media feeds. Despite having access to these data sources, investigators frequently struggle to locate hidden relationships among suspects because the information is highly fragmented.

Manual link analysis is slow, labor-intensive, and prone to missing critical, non-obvious connections. For example, a suspect might be linked to a crime scene indirectly through three phone calls to a middleman, followed by a bank transfer. Finding this connection manually across hundreds of pages of logs is nearly impossible in high-pressure environments.

Key Challenges in Current Workflows:

- **Data Fragmentation:** Information resides in siloed formats (PDF, Excel, scanned files).
- **Unstructured Text:** Valuable lead data is locked in narrative text (FIRs) in local Indian languages.
- **Relationship Visibility:** Lack of specialized tools to trace multi-hop connections between suspects.
- **Time Criticality:** Investigation delays allow suspect syndicates to adapt or escape.

Objective of the Proposed Solution:

To develop an AI-powered software platform that ingests heterogeneous documents, automatically parses and extracts entity relationships using Natural Language Processing (NLP), populates a high-performance **Graph Database (Neo4j)**, and offers investigators an intuitive, interactive link-analysis visualization to uncover organized syndicates.

2. Technical Architecture & Component Stack

The system is designed with a decoupled architecture. The backend is built using python for robust AI and graph processing, while the frontend is a web dashboard engineered for low latency and high-fidelity network renders.

Component	Technologies Used	Role / Purpose
Frontend UI	React / Tailwind CSS	Interactive dashboard for file uploads and searches.
Graph Render	Vis.js / Sigma.js / D3	Renders real-time interactive nodes & connections.
Backend API	FastAPI (Python)	Exposes async endpoints for ingestion, search, & calculations.
NLP Engine	spaCy / HuggingFace BERT / Ollama	Named Entity Recognition (NER) & relationship extraction.
Graph Database	Neo4j / NetworkX	Saves node networks and runs rapid graph traversals.
Data Store	PostgreSQL / SQLite	Stores raw file archives and user access logs.

Core Processing Pipeline:

- **Step 1 - Data Ingestion:** The investigator uploads files (FIR narrative PDFs, CDR Excel spreadsheets, Bank transfer CSVs).
- **Step 2 - NLP Processing:** An AI pipeline parses unstructured text, extracting entities (Suspect name, Phone number, Email, Address, Location, Bank account) and relations (called, transfer_to, associate_of).
- **Step 3 - Graph Construction:** Nodes and directed edges are created in Neo4j. Duplicate nodes are resolved based on shared attributes (e.g., same phone number merges two suspect nodes).
- **Step 4 - Network Analytics:** Algorithms run to identify clusters, compute node centrality, and trace communication paths.
- **Step 5 - Visual Analytics:** The frontend renders the network dynamically. Investigators can filter, expand, or highlight paths.

3. MVP Scope & Smart India Hackathon Roadmap

Proposed Modules for the 36-Hour Hackathon:

Module A: The NLP Extractor & Translator: A Python-based document reader. Since SIH evaluates real-world adaptation, this module will include a *Translation layer* using open-source translation models (e.g., HuggingFace MarianMT) to translate vernacular FIRs (e.g., Hindi) into English before extracting entities. This handles regional Indian data effectively.

Module B: Graph Analytics Engine: Leverages Neo4j (or local NetworkX) to perform complex operations:
1. *PageRank Centrality:* Measures who is the most influential node, exposing the ring leaders/kingpins.
2. *Dijkstra Shortest Path:* Calculates how Suspect A is connected to Suspect B through middlemen.
3. *Community Detection:* Groups nodes to identify separate operating gangs.

Module C: Interactive Investigator UI: A modern web dashboard that avoids static list grids. It uses a physics-simulated canvas where nodes can be dragged. Investigators can select two suspects and click 'Find Connection' to highlight the chain of communication in bright amber. It also includes a **Timeline Playback Slider** to watch the network build up chronologically (e.g., showing transaction flows over time).

Judges' Wow Factors (Features to Stand Out):

- **Timeline Playback:** A slider to visualize the network's evolution step-by-step. Judges love interactive temporal UI components.
- **Multi-Hop Trail Highlighting:** Automatically highlights financial money-laundering cycles (e.g. A to B, B to C, C to A).
- **Zero-Setup Fallback Mode:** The code should detect if Neo4j is offline and automatically fallback to an in-memory database. This ensures your live demo never crashes due to database connection errors.
- **Robust Mock Datasets:** Prepare high-quality simulated databases depicting a mock cyber fraud chain so the demo tells a clear, cohesive story.

Conclusion

This software-based architecture addresses the SIH26189 guidelines for the Ministry of Home Affairs. It provides actionable intelligence, streamlines coordination between jurisdictions, and is highly feasible for a dedicated team to develop into a working prototype within the 36-hour hackathon timeframe.